

Visual Grounding in Remote Sensing Images

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ABSTRACT

Ground object retrieval from a large-scale remote sensing image is very important for lots of applications. We present a novel problem of visual grounding in remote sensing images. Visual grounding aims to locate the particular objects (in the form of the bounding box or segmentation mask) in an image by a natural language expression. The task already exists in the computer vision community. However, existing benchmark datasets and methods mainly focus on natural images rather than remote sensing images. Compared with natural images, remote sensing images contain large-scale scenes and the geographical spatial information of ground objects (e.g., longitude, latitude). The existing method cannot deal with these challenges. In this paper, we collect a new visual grounding dataset, called RSVG, and design a new method, namely GeoVG. In particular, the proposed method consists of a language encoder, image encoder, and fusion module. The language encoder is used to learn numerical geospatial relations and represent a complex expression as a geospatial relation graph. The image encoder is applied to learn large-scale remote sensing scenes with adaptive region attention. The fusion module is used to fuse the text and image feature for visual grounding. We evaluate the proposed method by comparing it to the state-of-the-art methods on RSVG. Experiments show that our method outperforms the previous methods on the proposed datasets. <https://sunyuxi.github.io/publication/GeoVG>

CCS CONCEPTS

• **Information systems** → *Image search*; **Multimedia and multimodal retrieval**.

KEYWORDS

dataset, object retrieval, visual grounding, remote sensing, referring expression

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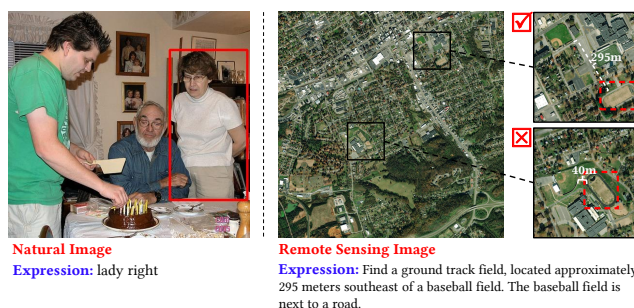


Figure 1: Two examples demonstrate the differences between visual grounding in natural images and visual grounding in remote sensing images. (1) remote sensing images (right) are geospatial images containing longitude and latitude, but natural images (left) are not. Correspondingly, the language expressions for remote sensing objects contain complex numerical geospatial relations such as distance. (2) remote sensing images are large-scale scenes and target objects are inconspicuous. However, natural images are small-scale scenes and target objects are generally salient.

1 INTRODUCTION

Remote sensing is the process of detecting and monitoring the physical characteristics of ground objects[28]. Some special cameras, such as on satellites or aircraft, are used to collect remotely sensed images. These remote sensing (RS) images usually cover a very broad area on the Earth's surface and contain a large number of objects, as shown on the right of Figure 1. These objects correspond to real-world objects (e.g., baseball fields, buildings). A variety of applications, such as environmental monitoring[12] and disaster assessment[3], need to detect and identify these objects. For example, in disaster management, it is necessary to retrieve some damaged objects from some large-scale scenes to deal with disasters. However, we cannot manually and quickly find some

objects of interest in large-scale scenes. Therefore, it is important to automatically localize a target object from large-scale scenes by some instructions, such as a language expression.

In this paper, we propose a novel problem for visual grounding in remote sensing images. Visual grounding aims to locate the particular objects (in the form of a bounding box or segmentation mask) in an image by a natural language expression, e.g., the man with sunglasses. The task already exists in the computer vision community. However, current benchmark datasets[2, 15, 16, 19, 29] and on top of which various models[10, 24, 35, 36] mainly focus on natural images, such as those captured by phone cameras, the remote sensing images are less considered.

The visual grounding on remote sensing images is different from the conventional visual grounding task. First, RS images contain the geographical coordinates (e.g., longitude, latitude) and spatial resolution of ground objects. Correspondingly, language expressions of remote sensing objects contain complex numerical geospatial relations and attributes, such as 100 meters north of, 200 square meters. These characteristics do not exist in the natural image domain. In particular, these numerical relations (attributes can be considered as special self-connected relations) are very important for localizing ground objects, and their representations need to be focused on. For example, as shown in Figure 1. An expression, like “find a ground track field, located approximately 295 meters southeast of a baseball field. The baseball field is next to a road.”, is provided. If we do not know the distance between the ground track field and the baseball field, the false object may be detected, as shown in the bottom right of Figure 1. Second, remote sensing images include larger areas of objects than natural images, and finding an object in remote sensing images is very difficult. For example, WorldView-1, a high-resolution optical satellite, captures images up to 750,000 square kilometers¹, and natural images typically show objects within 0.5 km². The large imaging range indicates that a lot of objects are contained in RS images and objects of interest are very small and densely clustered, rather than the large and prominent objects in natural images. Thus, in RS images, the target objects are not easily identified and the visual grounding is more challenging.

To the best of our knowledge, no existing datasets can be used to study visual grounding in remote sensing images. Hence, we propose a new dataset, namely RSVG, to evaluate this task. RSVG is featured in several aspects. First, it contains multi-spatial-resolution satellite or aerial images collected from different sensors, where each image covers an area ranging from 0.06 km² to 25 km². The max area is equivalent to one-twelfth the size of the Tokyo City limits² and is large enough for a lot of applications. Second, it contains the referred objects and corresponding sentences with a diverse set of numbers representing the area, distance, ordinality, etc. When selecting and describing an object in a remote sensing image, the area of the object and its distance from other ground objects need to be manually measured. This is very time-consuming and tends to exhibit strong biases, i.e., only salient objects are selected. Inspired by [2, 16, 17], we analyze the attributes and spatial relations of objects and summarize some text templates. Then the description of an object is generated with OpenStreetMap, an open-source

map that provides detailed ground object information, including categories, and boundaries. Similar to [19], we ask some people to use the description to localize the selected object. If the object cannot be accurately identified, the description will be discarded. Finally, we collected 4,329 remote sensing images with a total of 7,933 annotations of text expressions on 5,994 objects.

We evaluate several state-of-the-art visual grounding models (e.g., [10, 35, 36]) on our RSVG dataset, and find all of them get a large drop compared to their performance on natural images. First, existing methods cannot learn complex numerical spatial relations well when encoding text information. For example, these methods treat numbers as special words or sub-words, lose the smoothness of numbers, and cannot effectively learn their representations. Second, existing methods cannot effectively identify key regions and locate target objects in large-scale remote sensing scenes when encoding images.

To address these drawbacks, we propose a novel visual grounding method for geospatial images, namely GeoVG, which mainly contains three components: a language encoder with geospatial relations, an image encoder based on adaptive region attention, and a fusion module. The language encoder contains a numerical context module and represents an expression as a geospatial relation graph, where nodes and edges denote object and relation, respectively. The numerical module can effectively learn these number representations. The graph can clearly display the relations between multiply ground objects so that the complex expression content can be easily understood and learned. The image encoder adopts adaptive region attention to reduce the search range in large-scale scenes. The fusion module uses a self-attention network to identify some valid edges from the graph, and fuse these edges into adaptive visual content through a FiLM module. The valid edges imply a reasoning path of a target object. The accuracy of locating objects in large-scale scenes can be further improved after visual content fuses these valid edges.

The contributions of this paper can be summarised as follows: 1) We introduce a new challenging problem, i.e., visual grounding in remote sensing images. 2) We build a new dataset on top of large-scale remote sensing images, called RSVG. 3) We design a novel visual grounding solution for remote sensing images, namely GeoVG. Comprehensive experiments have been conducted to demonstrate that our GeoVG model significantly outperforms state-of-the-art methods on RSVG.

2 RELATED WORK

2.1 Datasets

We briefly describe existing visual grounding datasets. RefCOCO[11], RefCOCO+[11], and RefCOCOg[19] are collected on top of MSCOCO[13] images. The expressions and bounding boxes of RefCOCO and RefCOCO+ are generated through a two-player game. RefCOCOg is collected in a non-interactive setting by MTurk workers. These three datasets use a single expression to find objects. GuessWhat?![4] is another dataset based on MS-COCO images and uses a dialog, i.e. a sequence of sentences, to perform visual grounding. GuessWhat?! can only test the final results of models and does not evaluate their reasoning process. CLEVR-Ref+[16] is a synthetic diagnostic dataset built on the CLEVR environment. The CLEVR-Ref+ dataset

¹<https://en.wikipedia.org/wiki/WorldView-1>

²https://en.wikipedia.org/wiki/Greater_Tokyo_Area

can evaluate the intermediate reasoning process of models. However, the images in CLEVR-Ref+ are synthetic and cannot fully reflect real-life scenarios. Cops-Ref[2] is collected on top of real images and is applied to exploit the full reasoning chain embodied in an expression. Usually, an expression uses rich knowledge to describe a target object, and we need to combine some commonsense knowledge to understand it. KB-Ref[29] provides a dataset containing commonsense knowledge from Wikipedia, ConceptNet[26]. These datasets mentioned above are all about natural images. But, natural images are small-scale images without coordinate information, which fail to display extensive ground information. Remote sensing images can capture ground objects in a very wide range. Therefore, we propose a visual grounding dataset RSVG on remote sensing images.

2.2 Visual Grounding

Existing visual grounding methods can be categorized into two-stage and one-stage. Two-stage methods[5, 8, 9, 20, 21, 32–34, 37–40] need to generate candidate proposals using a pre-trained object detection network for an image. In remote sensing images, too many objects are obtained and two-stage methods are very inefficient. Moreover, there is no universal object detector that covers all ground objects[31]. Thus, we mainly focus on one-stage methods, which directly predict the bounding box. For example, OneStage[36] encodes an expression as a single vector and concatenates it at all locations of visual features. OneStage loses the details of long expressions. ReSC[35] proposes a recursive sub-query construction framework, where an expression is gradually divided into multiple sub-queries. ReSC can deal with long expressions, but it ignores the spatial relationship of objects in images. LBYLNet[10] applies a landmark features convolution to encode contextual information of an image. These methods all deal with natural images but remote sensing images. In the paper, we propose a new method GeoVG to locate an object in remote sensing images.

3 THE RSVG DATASET

Previous datasets[2, 15, 16, 19, 29] of visual grounding typically only contain small-scale natural images without geospatial information. In this paper, we collect a new dataset called RSVG that contains large-scale remote sensing images. Compared with previous datasets, RSVG has two distinct characteristics, namely 1) the contained remote sensing images are geospatial images, which have a geographical coordinate and spatial resolution, and 2) the contained scenes are usually large-scale. Figure 1 on the right shows a typical example of our dataset. In this section, we will describe our data collection pipeline (including image collection in Section 3.1, expression generation in Section 3.2) and give a statistical analysis of the dataset in Section 3.3.

3.1 Image Collection

RSVG mainly contains remote sensing images, which consist of satellite and aerial images. These images are collected from multiple sensors and platforms (e.g. Google Earth) with multiple resolutions. To increase the diversity of the data, we collect images taken in several cities in the United States and record the exact geographical

coordinates and spatial resolutions of these images. These coordinates and resolutions are very important to measure distance, area, and spatial relation between ground objects. The size of these images is 1024×1024 pixels and their coverage varies from 0.06 km^2 to 25 km^2 . The coverage is large enough for a lot of applications. In the end, we obtained 7,699 remote sensing images.

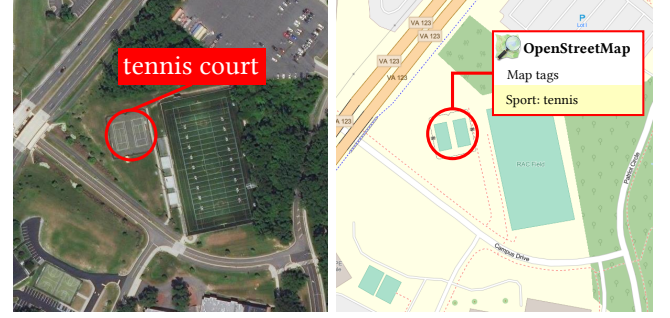


Figure 2: An example of OpenStreetMap data.

3.2 Expression Generation

RSVG consists of an image, a bounding box of the target object, and a referential expression that uniquely locates the target object. The bounding box and expressions are manually annotated. In RSVG, the image is a large scene and the target object is not easily found. In addition, the expressions contain a lot of numerical distance and area information about ground objects. These values need to be manually measured, which is very time-consuming. We take inspiration from [2, 16], in which the authors build a visual grounding dataset about synthetic images following an automated procedure. We also design some referring expression templates with slots to generate descriptions. The slots mainly involve the category and attributes of objects and the spatial relation between them. Existing methods aim at synthetic images, and the slots are easily filled with some extracted information (e.g., category and color) from synthetic images. However, RSVG is interested in real remote sensing images, and we cannot automatically obtain the missing data in the template. Thus, we leverage the openly accessible OpenStreetMap containing ground object information (e.g., category and location) provided by volunteers. In this section, we first introduce the category of ground objects and their attributes extracted from OpenStreetMap. Then we depict and analyze the spatial relation between grounding objects with these extracted data. Next, we design some referring expression templates. After that, the templates are filled with these extracted data to generate expressions. Finally, we ask some people to check these expressions.

Ground Object Category. OpenStreetMap represents geographic attributes (e.g., geo-location) of ground objects using various tags. These geographic data are contributed by lots of volunteers. To obtain these crowdsourced data, we first localize each image in OpenStreetMap with coordinates of its four corners. Then we parse the properties of scenes present in the corresponding region. As shown in Figure 2, two tennis courts are annotated in OpenStreetMap.

Index	Forms	Exemplar templates	Expression examples
1	Zero-reference-based	The <attr0> <obj0> <rel0>.	The largest baseball field.
2	One-reference-based	The <att0> <obj0> <rel0> a/an <att1> <refobj1>.	The swimming pool located approximately 158 meters southeast of a tennis court.
3	Two-references-based	The <att0> <obj0> <rel0> a/an <att1> <refobj1> and a/an <att2> <refobj2>.	The baseball field in the middle of a blue build and a track field.
4	Three-references-based	The <att0> <obj0> <rel0> a/an <att1> <refobj1>. Within <rel1> of the <att1> <refobj1>, there are a/an <att2> <refobj2> and a/an <att3> <refobj3>.	The baseball field located approximately 200 meters southeast of a tennis court. Within 1000 meters of the tennis court, there are a blue build and a track field.

Table 1: Examples of referring expression templates. *obj0* denotes a target object or multiple adjacent target objects. *refobj_{1,2,3}* represents one or more reference objects. *attr_{0,1,2,3}* and *rel_{0,1,2,3}* denote the corresponding attributes and relations, respectively.

In this way, we automatically obtain lots of ground objects with annotated attributes, including geo-location, boundary, area, color, shape, and distance. In particular, not all objects are considered. We mainly pay attention to some common objects, including baseball field, basketball court, tennis court, football field, ground track field, swimming pool, storage tank, water tower, wastewater plant, bridge, building, roundabout, river, lake, pond, parking lot, road, and tree. These categories are usually used by existing object detection datasets [31]. These accessible data are all basic data of objects. The complex attributes, such as spatial relation, cannot be obtained directly from OpenStreetMap, and we need to measure them based on geo-location, boundary, etc.

Spatial Relations. The spatial relation refers to the way objects are arranged in relation to one another in geographic space, and the relationship is very critical to find a target object in a visual grounding task. For example, a referring expression is that a U-shaped white building is located approximately 1000 meters north of a track field. In the expression, the word “north” represents the directional relation between the building and the track field. If “north” is not involved, we may find a building in other directions such as east, or west. Besides, the spatial relations used in remote sensing images are very different from the spatial relations in natural images. For example, left, right, etc. are often applied to natural images. In remote sensing images, we tend to use east, west, etc., and joint some numerical distances to depict the spatial relations of objects. Current methods usually fail to learn spatial relations with numerical distances.

As suggested by [18], the spatial relations are categorized into topological relations, directional relations, distance relations, and n-ary spatial relations.

- **Topological relations** are generally defined as the spatial relationships between adjacent or neighboring ground objects. For example, find the baseball field next to a track field.
- **Directional relations** are known as cardinal directions. They are qualitative spatial relations that describe how an object is placed relative to other objects. The common directions include east, west, south, north, southeast, southwest, northeast, and northwest. For example, find the baseball field which is located to the north of a track field.

- **Distance relations** are also called spatial proximity. It depicts how far is the object away from the reference object. For example, find a baseball field within 1000 meters of a track field.
- **N-ary relations** depict the spatial relations between/among (two or more) ground objects. For example, find the baseball field in the middle of two track fields.

These spatial relations can be obtained based on the extracted geographic attributes from OpenStreetMap. For example, given the geo-location of two objects, we can measure their distance. If the distance is less than a certain threshold, we regard them as next to each other, i.e., obtain their topological relations.

Referring Expression Template. The expression template is the key to construct our dataset, responsible for generating grammatically correct and unambiguous expressions for each of the described target objects. As we know, we usually uniquely locate a target object by describing the target object, its surrounding reference objects, and the spatial relationship between them. For example, find the U-shaped white building, which is located to the north of a track field. The building is a target object. The track field is a reference object. “North” indicates the spatial relation. This expression involves only one reference and is very simple. When locating objects in complex environments, we may consider more reference objects. Therefore, expression templates can be designed according to these objects and relationships.

In the paper, we only focus on one target object and multiple adjacent objects are also considered as one. As shown in Table 1, the designed templates can be categorized into four types according to the number of reference objects: zero-reference-based, one-reference-based, two-references-based, and three-references-based templates. The zero-reference-based template only refers to a target object, such as the <attr0> <obj0> <rel0> (the largest baseball field). The terms “attr0”, “obj0” and “rel0” represent an attribute, a ground object and a spatial relation. The one-reference-based template means that the target object is located by relying on one reference object. For example, the <att0> <obj0> <rel0> a/an <att1> <refobj1> (the swimming pool located approximately 158 meters southeast of a tennis court). The term “refobj1” means a reference object. Other templates are also similar. Usually, templates involving three references are very complex and are able to satisfy most practical applications. These designed basic templates can also be

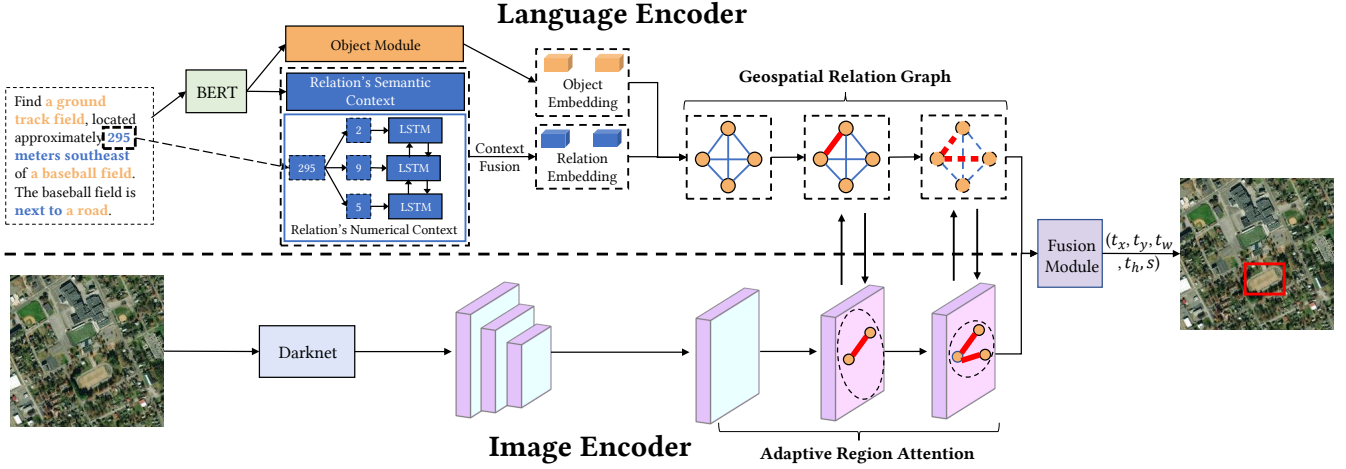


Figure 3: The overall architecture of our baseline model GeoVG, which contains three components: a language encoder with geospatial relations, an image encoder based on adaptive region attention, and a fusion module. These modules are described in Section 4.

Dataset	Referring expressions	Object instances	Images	Average expression length
total	7933	5994	4239	28.41
training	5505	4183	2963	28.33
validation	1201	903	636	29.09
test	1227	908	640	28.10

Table 2: Statistics of RSVG dataset.

further composed with each other to generate more complex expressions. In addition, different spatial relationships are selected in these templates according to the number of references. For example, if the reference is one, the N-ary relation is not suitable and the other relations can be selected.

Expression Generation and Post Processing. With the ready expression templates, the referring expression can be easily produced by filling the predefined templates with the content from OpenStreetMap. For example, given an expression template like the $\langle att0 \rangle \langle obj0 \rangle \langle rel0 \rangle a/an \langle att1 \rangle \langle refobj1 \rangle$, we can generate “the swimming pool located approximately 158 meters southeast of a tennis court” for a certain remote sensing image. The filled object is extracted automatically from the image. Since we know the exact geo-location and properties of objects in the images, we can project them back to the image plane to get the ground truth bounding box. In this way, crowdsourced descriptions and annotations of target objects in remote sensing images can be automatically yielded at a very low cost compared to conventional manual labeling.

After getting these descriptions and bounding boxes, we still need some post-processing operations and checks, because OpenStreetMap may provide some wrong data. For example, some ground objects were removed, but the map was not updated in time. Some buildings are marked as basketball courts, but we cannot identify them based only on remote sensing images. To deal with these incorrect data, we first ask some people to check whether the extracted categories and attributes are correct. Then, others are presented

with the images and referring expressions, and are asked to click inside the objects being referred to. If the clicked points are inside the bounding boxes of target objects, we consider the expressions valid and keep them, otherwise we discard them. Besides, as mentioned in [16], when there is only one instance of the category of the target object in the image, the model will exhibit a strong bias, i.e. only recognizing the category and discarding the referring expression also yield a much higher baseline. To eliminate bias, we abandon those images that contain only one instance of the target category. After this post-processing, we obtain a clean and accurate dataset.

3.3 Statistics of the Dataset

After the above processing, we collected 4239 remote sensing images with a total of 7933 annotations of text expressions on 5994 objects, as shown in the first row of Table 2. In the dataset, the average number of referring expressions in each image is 1.87. The average length of the expressions is 28.41 words. We randomly split the dataset on the base of the images for training, validation, and test. There are 5505/1201/1227 expressions for training/validation/test, respectively, and the last three rows in Table 2 show the data distribution.

4 METHOD

In the visual grounding task, given a natural language expression and a remote sensing image, the target is to locate the described object from the image. In the expression, ground objects and geospatial relations are usually involved. Geospatial relations include not only words, but also numbers that provide very critical information such as distance and area. These contents need to be accurately represented when the target object is localized. Existing methods treat numbers as special words or sub-words, lose the smoothness of numbers[27], and cannot effectively learn their representations. Moreover, these relations involve multiple ground objects and are very complex. Existing models also cannot effectively represent geospatial relations in a complex expression. In addition, remote

sensing images are large-scale scenes, and the target objects are difficult to identify in this long-range visual context. To address these drawbacks, we propose a novel visual grounding solution for geospatial images, namely GeoVG. The overall architecture is illustrated in Figure 3. GeoVG mainly contains three components: a language encoder with geospatial relations, an image encoder based on adaptive region attention, and a fusion module.

The language encoder constructs a geospatial relation graph for an expression, in which all objects are regarded as nodes and edges are the relations generated by fusing semantic and numerical context. In particular, a numerical context module is designed to represent these numbers. A complex expression is explicitly modeled as a graph. The contained objects and relations are easier to learn. The constructed graph is a fully connected graph and not all edges are valid. Therefore, we cannot apply it directly to locate an object in an image and need to further use image content to identify valid edges. The image encoder uses an adaptive region attention network to select the regions of interest, where target objects and reference objects are contained with high probability. The adaptive network effectively reduces the search range in large-scale scenes. The fusion module uses the selected visual content to identify valid edges through a self-attention network in the geospatial relation graph. After that, these edges are fused into visual content through a FiLM module[22] and final feature maps are produced to predict the bounding box of a target object. Since these valid edges contain reasoning paths, the difficulty of finding target objects in large scenes is further reduced by fusing these paths.

4.1 Language Encoder with Numerical Geospatial Relations

The language encoder mainly constructs a geospatial relation graph, where nodes and edges correspond to ground objects and geospatial relations. Each edge that connects two objects is generated through a dot attention network, where these two objects are used as queries and all relations in the expression are keys and values.

First, we introduce node generation. Similar to [35], each word is encoded into a vector w_i , where i denotes the i -th word in the expression. We additionally obtain the part of speech (POS) of each word, i.e., t_i , by an open source POS tagging software[23]. All nouns are regarded as objects, including a target object and reference objects. We design an object module to generate embeddings as follows:

$$v_k = \text{Norm}(\text{Conv}(\text{MLP}(w_i) \parallel t_i)), \quad (1)$$

where MLP is a two-layer linear network, and t_i equals 1 if w_i is a noun, otherwise 0. $\parallel$ concatenates the embedding and POS data to select possible ground objects. Conv and Norm denote a convolution network and a normalization layer, respectively. Each concatenated data is treated as a channel and the convolution transforms these concatenated data into object embedding v_k . A total of K embeddings are generated. In this way, we generate K nodes.

Next, we produce all geospatial relation embeddings. Each expression usually includes multiple geospatial relations and each relation contains word and number contents, corresponding semantic context, and numerical context. In particular, a numerical context module is designed to learn these numbers' representations,

as follows:

$$\begin{aligned} x_i^c &= W_c^T c_i, \\ h_{i+1}^c &= \text{BiLSTM}(x_i^c, h_i^c), \end{aligned} \quad (2)$$

where c_i is the one-hot vector of the i -th char in a number, W_c^T denotes a trainable lookup table, h_i is the hidden state at time step i , and BiLSTM is used to learn the representations of a number. The last states are treated as the relation's numerical context embedding e_i^c . In addition, the relation's semantic context embedding e_i^s is produced by BERT. To fuse the two contexts, a convolution is applied as follows:

$$e_k^r = \text{Norm}(\text{CONV}(e_i^s \parallel e_i^c \parallel t_i)), \quad (3)$$

where t_i equals 1 if the raw word is a number, adjective, or adverb. In this way, a relation embedding e_k^r is obtained and the total number is $K \times K$.

Finally, each edge is constructed by a dot product attention, where we use its associated objects as the queries as follows:

$$v_{ij}^q = \text{ReLU}(W_v^T (v_i \parallel v_j) + b_v).$$

All geospatial relation embeddings are regarded as keys and values. The attention is formulated as:

$$\begin{aligned} s_{ij} &= \frac{\exp(v_{ij}^q e_{ij}^r)}{\sum_{m,n} \exp(v_{ij}^q e_{m,n}^r)}, \\ e_{ij} &= \sum_k s_{ik} e_{kj}^r, \end{aligned} \quad (4)$$

where s_{ij} is an attention weight and e_{ij} is a generated edge for nodes v_i and v_j . In this way, a geospatial relation graph is constructed. But, the graph is a fully connected graph, and not all edges are important and valid. In the fusion module, we employ some visual features to select key edges.

4.2 Image Encoder based on Adaptive Multi-region Attention

Remote sensing images are usually large-scale, but ground objects are very small. The objects or relations of interest are not easy to find. We introduce an adaptive region attention module that distills key visual content from a large-scale scene, which is formulated as:

$$\begin{aligned} g_i^l &= f_i^l s_i^l, \\ s_i^{l+1} &= \sigma(\text{CONV}(W_g^T g_i^l + b_g)), \\ f_i^{l+1} &= s_i^{l+1} f_i^l, \end{aligned} \quad (5)$$

where f^* and s^* denote visual features and attention scores. In particular, the image is also used to select some valid edges. For each selected edge, we execute the attention module once. f^l and s^l denote visual features and attention scores when the l -th edge is selected. s^0 is 1 and f^0 denotes raw visual features extracted by DarkNet-53[25].

4.3 Fusion Module

In this section, we mainly fuse the constructed geospatial relation graph into visual features. The fused visual features can be used to predict the bounding box of a target object.

First, we design an attention network to select some valid edges based on visual features. A total of L edges are selected. The attention is formulated as:

$$\begin{aligned} g_{ij}^l &= \text{MLP}(e_{ij} \parallel v_i \parallel v_j), \\ \alpha_{ij}^l &= \text{CONV}(\beta_{ij}^{l-1} \odot g_{ij}^l \odot \text{mean}(F_{i+j}^{l-1})), \\ \lambda_{ij}^l &= \frac{\exp(\alpha_{ij}^l t)}{\sum_{m,n} \exp(\alpha_{m,n}^l t)}, \\ e_v^l &= \sum_{i,j} \lambda_{ij}^l g_{ij}^l, \end{aligned} \quad (6)$$

where $\odot$ is an element-wise multiplication and F^{l-1} denotes the visual feature that fuses the selected $l-1$ valid edges. β_{ij}^{l-1} is a filter to avoid an edge being selected repeatedly. β_{ij}^{l-1} is close to 0 if e_{ij} is a valid edge, otherwise approximately equal to 1. $\beta^0 = \mathbf{1}$ and we let $\beta^l = 1 - \lambda^{l-1}$. λ_{ij}^l is an attention score that is generated by a parameterized softmax, where t is a temperature parameter. When t takes a suitable value, the softmax can make one value of λ^l approximate 1 and the other values close to 0. e_v^l is the l -th valid edge we need to generate. The remaining valid edges are also obtained similarly.

Next, we fuse these valid edges into visual features through a FiLM module[22], which is formulated as:

$$\begin{aligned} \gamma_0^l &= \text{MLP}_0(e_v^l), \quad \gamma_1^l = \text{MLP}_1(e_v^l), \\ F^l &= \text{ReLU}(\text{Conv}(\text{ReLU}(\gamma_0^l \odot \text{Conv}(f^l \oplus Z) \oplus \gamma_1^l))), \end{aligned} \quad (7)$$

where $\oplus$ denotes the element-wise addition. Z represents the coordinate features with a 10-dimensional position embedding vector, such as latitude, longitude, and ground sample distance. The coordinate features are generated in a similar way to prior works[35, 36]. F^l is the visual features fusing valid edge information.

Finally, after valid edges are obtained and fused completely, the last visual features contain the reasoning path of the target object and also fuse the expression content. The visual features, i.e. feature maps, can be used to detect the bounding box of the target object. We adopt an anchor-based box regression head in YOLOv3 as a detection head. The loss function is the same as[36].

5 EXPERIMENT

5.1 Experimental Settings

Implementation Details. Following LBYLNet[10] and ReSC[35], we extract visual features by Darknet-53 pre-trained on COCO object detection[13]. Since lots of small objects are contained in remote sensing images, we keep the original image size, i.e., 1024×1024 . The words of expressions are encoded as a 768D vector with the uncased base version of BERT[6, 30]. Expressions with words less than 80 are padded. In this paper, we have two hyper-parameters K and L , which represent the number of objects and the number of valid edges, respectively. We empirically set $K = 5$ and $L = 3$. Besides, we generate the anchor boxes by K-means clustering following the procedure of [25]. Our proposed method is implemented with the Pytorch framework. The RMSProp optimizer is used to train the model with an initial learning rate of 0.0001. The learning rate decreases by half every 25 epochs. The batch size is 4 in all

our experiments and the total number of epochs is 50 for the RSVG dataset.

Evaluation Metric. We follow [15] to use accuracy $\text{acc}@0.5$ and $\text{acc}@0.25$ as our evaluation metric. The accuracy is calculated by checking whether the bounding boxes of target objects are correctly predicted or not. The highest scored bounding box from our network is regarded as the predicted box of a target object. $\text{acc}@0.5$ means that the IoU of the predicted box is at least 0.5 with the ground truth bounding box. $\text{acc}@0.25$ is also defined similarly.

Table 3: Performance (acc%) comparison with state-of-the-art approaches and our proposed GeoVG on RSVG.

Method	Test		Validate	
	acc@0.5	acc@0.25	acc@0.5	acc@0.25
NMTree	14.26	18.42	12.91	17.65
MAttNet	21.19	27.30	19.90	26.89
Ref-NMS+NMTree	24.12	30.81	21.48	27.89
OneStage	30.15	34.06	30.06	34.47
LBYLNet	32.19	35.21	31.64	36.47
ReSC	51.18	57.05	53.96	58.45
GeoVG	59.40	64.95	58.20	65.11

5.2 Comparison with State-of-the-art

In the paper, we compare the proposed method with two-stage methods (e.g., NMTree[14], MAttNet[37], and Ref-NMS+NMTree[1]) and one-stage methods (e.g., OneStage[36], ReSC[35], and LBYLNet[10]). In the two-stage methods, we adopt RoITransformer[7] to extract object proposals and their visual features. The one-stage methods are trained from scratch on the training split of our proposed RSVG dataset, using their own training strategies.

Performance of the Visual Grounding Models. The accuracy of all the baselines and the proposed method is reported in Table 3. From the results, we can observe that most one-stage methods outperform two-stage methods. This is because there is no universal pre-trained object detector that covers all ground objects, such as buildings. In the one-stage methods, OneStage[36] obtains the worst performance because it can neither deal with the complex expression with geospatial relations nor large-scale image scene content well. LBYLNet performs slightly better than OneStage since it applies a landmark feature convolution to encode contextual information of the large-scale image. Compared with LBYLNet, ReSC obtains a marginal improvement by a recursive sub-query construction framework. This may be because ReSC splits expressions into multiple sub-queries recursively, which is better able to learn complex spatial relationships and fuse them into images. Our proposed method GeoVG yields better performance than all the baselines. For example, in the test dataset, compared with ReSC, GeoVG achieves 16.06%, 13.85% increment on $\text{acc}@0.5$ and

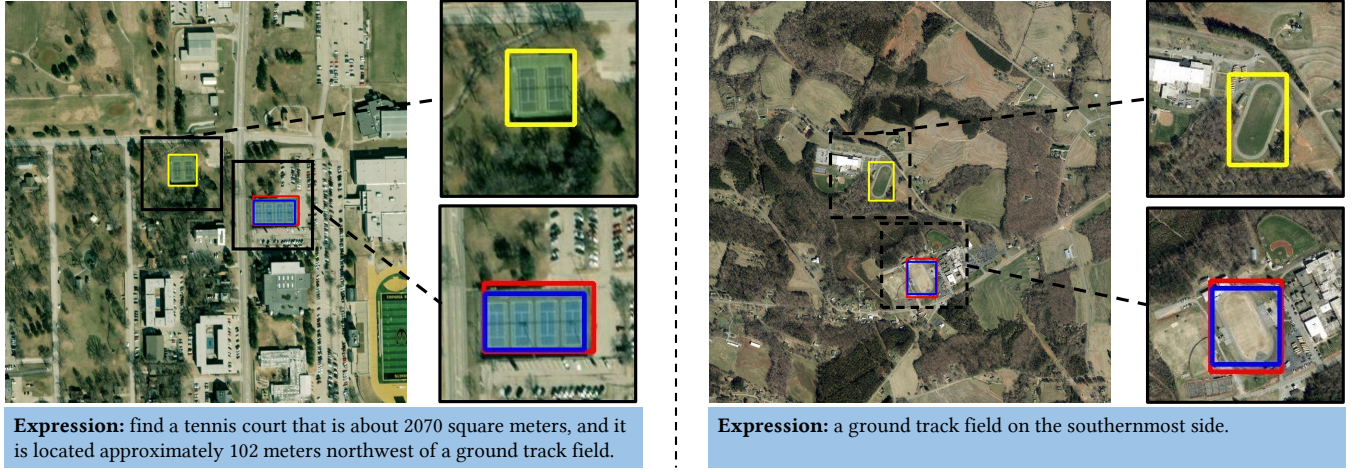


Figure 4: Visualization of some experimental results. Ground-truth bounding boxes (in red) are presented in the image. The blue one is the object selected by GeoVG. And the yellow one is chosen by ReSC.

Table 4: Ablation study on GeoVG

Method	Test		Validate	
	acc@0.5	acc@0.25	acc@0.5	acc@0.25
w/o numerical context	55.23	59.97	56.62	62.36
w/o adaptive attention	58.01	63.81	57.53	64.44
GeoVG	59.40	64.95	58.20	65.11

acc@0.25, respectively. GeoVG also produces similar improvements in the validation dataset.

The advantage of GeoVG comes from its two characteristics: (i) we develop a geospatial relation encoder to fully represent the complex expression, (ii) a image encoder and fusion module is proposed to deal with large-scale remote sensing images based on adaptive region attention. To illustrate the superiority of GeoVG, we visualize the results of ReSC and GeoVG in Figure 4. As shown on the left of Figure 4, the expression includes the area of a tennis court, such as 2070 square meters, and the distance between a tennis court and a ground track field. ReSC fails to learn these data accurately and to locate a false tennis court. GeoVG can effectively represent these numerical geospatial relations and locate the target object successfully. As shown on the right of Figure 4, there are multiple ground track fields. GeoVG can use adaptive attention to discover key regions in a large image and utilize a fusion module to efficiently find the target object from global to local. However, ReSC cannot deal with large-scale scenes and fails to locate the target object. In conclusion, GeoVG has better performance than other methods in visual grounding in remote sensing images.

5.3 Ablation Study

To investigate the impact of different design choices in our proposed approach, we perform the ablation test by discarding the numerical

context module (denoted as w/o numerical context) and adaptive region attention module (denoted as w/o adaptive attention), respectively. Specifically, for w/o numerical context, the BiLSTM model and numerical context are removed directly and we use the original bert output to learn numerical spatial relation. For the w/o adaptive attention, we use the raw visual feature maps as input to fuse text expression. The ablation results are summarized in Table 4. From the results, we observe that both the numerical context module and adaptive attention module make great improvements to our GeoVG method. For example, in the test dataset, compared with the w/o numerical-context-based method, the GeoVG method achieves 7.55%, 8.30% increment on acc@0.5 and acc@0.25, respectively. Our method, compared with the w/o adaptive-attention-based method, also produces 2.40% and 1.79% increments on acc@0.5 and acc@0.25. Thus, both the numerical context module and adaptive region attention module can help GeoVG to improve the accuracy of object localization under complex expressions and images.

6 CONCLUSION

In this work, we present a visual grounding dataset about remote sensing images, RSVG, in which images are large-scale geospatial images, and language expressions contain multiple ground objects and complex numerical geospatial relations. The existing method cannot effectively represent numerical geospatial relations and deal with large-scale scenes. We propose a novel visual grounding solution for geospatial images, namely GeoVG, which consists of a language encoder with geospatial relations, an image encoder based on adaptive region attention, and a fusion module. Experiments show that our proposed model indeed improves the performance on RSVG by a large margin.

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